Linux Networking Commands - Study Notes

1. Network Interfaces

Overview

- Network interfaces link the software side of networking to the hardware side
- Common interface names:
  - `eth0` - First Ethernet card
  - `wlan0` - Wireless interface
  - `lo` - Loopback interface (represents your computer, loops back to yourself)

The ifconfig Command

View all interfaces:

```
ifconfig -a
```

Example output:

```
eth0      Link encap:Ethernet  HWaddr 1d:3a:32:24:4d:ce
          inet addr:192.168.1.129  Bcast:192.168.1.255  Mask:255.255.255.0
          inet6 addr: fd60::21c:29ff:fe63:5cdc/64 Scope:Link
```

Key information in output:

- **HWaddr** - MAC address of the interface
- **inet addr** - IPv4 address
- **inet6** - IPv6 address
- **Mask** - Subnet mask
- **Bcast** - Broadcast address

Create and configure an interface:

```
ifconfig eth0 192.168.2.1 netmask 255.255.255.0 up
```

Bring interfaces up/down:

```
ifup eth0
ifdown eth0
```

The ip Command

Show interface information:

```
ip link show
```

Show interface statistics:

```
ip -s link show eth0
```

Show IP addresses:

```
ip address show
```

Bring interfaces up/down:

```
ip link set eth0 up
ip link set eth0 down
```

Add IP address to interface:

```
ip address add 192.168.1.1/24 dev eth0
```

2. Routing (route command)

Managing Routes with route Command

Add a new route:

```
sudo route add -net 192.168.2.1/23 gw 10.11.12.3
```

Delete a route:

```
sudo route del -net 192.168.2.1/23
```

Managing Routes with ip Command

Add a route:

```
ip route add 192.168.2.1/23 via 10.11.12.3
```

Delete a route:

```
ip route delete 192.168.2.1/23 via 10.11.12.3
```

or

```
ip route delete 192.168.2.1/23
```

3. DHCP Client (dhclient)

DHCP Client Overview

- dhclient starts on boot and configures network interfaces using DHCP
- Gets network interface list from `dhclient.conf`
- Tracks leases in `dhclient.leases` file across reboots

Obtain fresh IP address:

```
sudo dhclient
```

4. Network Manager

Network Manager Overview

- NetworkManager daemon automatically configures networks
- Appears as GUI applet on desktop taskbar
- Gathers hardware info on startup and activates connections

nm-tool Command

Check NetworkManager state:

```
nm-tool
```

Example output:

```
NetworkManager Tool

State: connected (global)

- Device: eth0  [Wired connection 1] ---------------------------------------
----
  Type:                 Wired
  Driver:               pcnet32
  State:                connected
  Default:              yes
  HW Address:           12:3D:45:56:7D:CC

  Capabilities:
    Carrier Detect:     yes

  Wired Properties
    Carrier:            on

  IPv4 Settings:
    Address:            192.168.22.1
    Prefix:             24 (255.255.255.0)
    Gateway:            192.168.22.2
    DNS:                192.168.22.2
```

nmcli Command

- Command-line tool to control and modify NetworkManager
- Refer to manpage for detailed usage

5. ARP (Address Resolution Protocol)

ARP Overview

- ARP cache stores IP-to-MAC address mappings
- Cache is empty on boot, populated as packets are sent

ARP Process

1. Source host creates Ethernet frame with ARP request
2. Source host broadcasts frame to entire network
3. Host with correct MAC address sends reply packet

4. Source host adds IP-to-MAC mapping to ARP cache
5. Packet transmission proceeds

View ARP Cache

Using arp command:

```
arp
```

Example output:

```
Address                  HWtype  HWaddress           Flags Mask
Iface
192.168.22.1             ether   00:12:24:fc:12:cc   C
eth0
192.168.22.254           ether   00:12:45:f2:84:64   C
eth0
```

Using ip command:

```
ip neighbour show
```

Key Configuration Files

- `/etc/network/interfaces` - Interface configuration
- `dhclient.conf` - DHCP client configuration
- `dhclient.leases` - DHCP lease tracking

Quick Reference Summary

Task	ifconfig/route	ip command
Show interfaces	`ifconfig -a`	`ip link show`
Show IP addresses	`ifconfig`	`ip address show`
Bring interface up	`ifup eth0`	`ip link set eth0 up`
Add IP to interface	`ifconfig eth0 192.168.1.1/24`	`ip address add 192.168.1.1/24 dev eth0`

Task	ifconfig/route	ip command
Add route	`route add -net 192.168.2.1/23 gw 10.11.12.3`	`ip route add 192.168.2.1/23 via 10.11.12.3`
Delete route	`route del -net 192.168.2.1/23`	`ip route delete 192.168.2.1/23`